MONOCHROMATIC SUBLATTICES IN RANDOM AND CARDINALITY COLOURINGS OF THE BOOLEAN LATTICE

DEEP BHATTACHARJEE

Abstract. Let $f(n)$ be the largest m such that every 2-colouring of the subsets of an n-element set has a monochromatic sublattice with m members. In a uniformly random k-colouring the largest monochromatic sublattice has $(1+o(1))\,n\log_k n$ members with high probability, so $f(n)\le(1+o(1))\,n\log_2 n$. We also find $f(n)$ for $n\le 22$ and $n=24$.

1. Introduction

A *sublattice* of $2^{[n]}$ is a nonempty family of subsets of $[n]=\{1,\dots,n\}$ that is closed under union and intersection. Erdős and Ulam asked how large a monochromatic sublattice every 2-colouring of $2^{[n]}$ must contain, and the same question for families closed under union only [Bl26, Problem #1183]. For $k\ge 2$ let $f_k(n)$ be the largest m such that every k-colouring of $2^{[n]}$ has a monochromatic sublattice with at least m members, and write $f(n)=f_2(n)$. A maximal chain $\varnothing=S_0\subset S_1\subset\cdots\subset S_n=[n]$ is a sublattice, and so is any subfamily of it, which gives

$$f_k(n)\ \ge\ \left\lceil\frac{n+1}{k}\right\rceil. \tag{1.1}$$

Erdős wrote that he had no plausible conjecture for the order of magnitude of $f(n)$ [Bl26].

The present state of the problem is the work of Bhattacharjee, Mandal and Bhattacharya [BMB26a, BMB26b], which settles both conjectures of Erdős and Ulam about union-closed families and proves, for sublattices,

$$\frac{11n}{20}\ \le\ f(n)\ \le\ (2+o(1))\,n\log_2 n. \tag{1.2}$$

The lower bound improves (1.1); it rests on a finite computation over a window of twenty levels [BMB26b, Theorem 7.6]. Chojecki [Ch26] had earlier obtained $f(n)\le(3+o(1))\,n\log_2 n$. The upper bounds come from a uniformly

2020 *Mathematics Subject Classification.* Primary 05D10; Secondary 06D05, 05A16, 05C80.

Key words and phrases. Boolean lattice, sublattices, Ramsey theory, random colourings, distributive lattices, Erdős problems.

random colouring, together with a count of the sublattices of $2^{[n]}$. Whether $f(n) = O(n)$ is open [BMB26b, Question 9.3].

This paper contains two results. The first determines the largest monochromatic sublattice of a random colouring to first order.

Theorem 1.1. *Let $k \geq 2$ be fixed and let χ be a uniformly random k-colouring of $2^{[n]}$. Let $\Lambda(\chi)$ be the largest number of members of a monochromatic sublattice. For every $\varepsilon > 0$,*

$$\mathbb{P}\big[(1-\varepsilon)\, n \log_k n \;\leq\; \Lambda(\chi) \;\leq\; (1+\varepsilon)\, n \log_k n\big] \;\longrightarrow\; 1 \qquad (n \to \infty).$$

Moreover, with probability tending to 1 every colour has a monochromatic sublattice with at least $(1-\varepsilon)\, n \log_k n$ members.

Corollary 1.2. *$f_k(n) \leq (1+o(1))\, n \log_k n$ for every fixed $k \geq 2$. In particular $f(n) \leq (1+o(1))\, n \log_2 n$.*

The upper bound halves the constant in (1.2). The lower bound in Theorem 1.1 shows that this is the limit of the method: any bound $f(n) \leq c\, n \log_2 n$ with $c < 1$, and in particular any proof that $f(n) = O(n)$, needs colourings far from random. The upper bound rests on a count of partial orders by their number of down-sets (Proposition 3.3), which is sharper than the counts used in [BMB26b, Ch26] in the range that matters: a sublattice with m members corresponds to a poset with m down-sets, and we show that there are only $2^{o(m)}$ naturally labelled posets on at most n points with $m \geq n \log_2 n$ down-sets. The lower bound is a product of two chains, one of them built greedily inside the random colouring (Figure 3).

The second result concerns small n. A colouring of $2^{[n]}$ is a *cardinality colouring* if the colour of a set depends only on its size, so that it is given by a word $g\colon \{0,1,\dots,n\} \to \{\mathsf{R},\mathsf{B}\}$. Write $f_{\mathrm{card}}(n)$ for the least possible size of the largest monochromatic sublattice of a cardinality colouring, so that $f(n) \leq f_{\mathrm{card}}(n)$. Howorka's theorem on union-closed families, recorded by Erdős, concerns these colourings, and they are natural candidates for good colourings of $2^{[n]}$: a sublattice coloured by its sizes behaves like a weighted poset whose down-set sums avoid one colour class (Section 5).

Theorem 1.3. *The values of $f_{\mathrm{card}}(n)$ for $1 \leq n \leq 25$ are*

$$1, 2, 2, 3, 3, 4, 4, 5, 5, 6, 7, 7, 8, 8, 9, 9, 10, 10, 11, 12, 12, 13, 14, 14, 15.$$

Combined with the lower bounds of [BMB26b], the upper bounds $f(n) \leq f_{\mathrm{card}}(n)$ settle $f(n)$ for most $n \leq 25$.

Theorem 1.4. *For $1 \leq n \leq 22$ and for $n = 24$,*

$$f(n) = f_{\mathrm{card}}(n).$$

Thus $f(n) = \lceil (n+1)/2 \rceil$ for $n \leq 10$ and for $n \in \{12, 14, 16, 18\}$, while $f(11) = 7$, $f(13) = 8$, $f(15) = 9$, $f(17) = 10$, $f(19) = 11$, $f(20) = 12$,

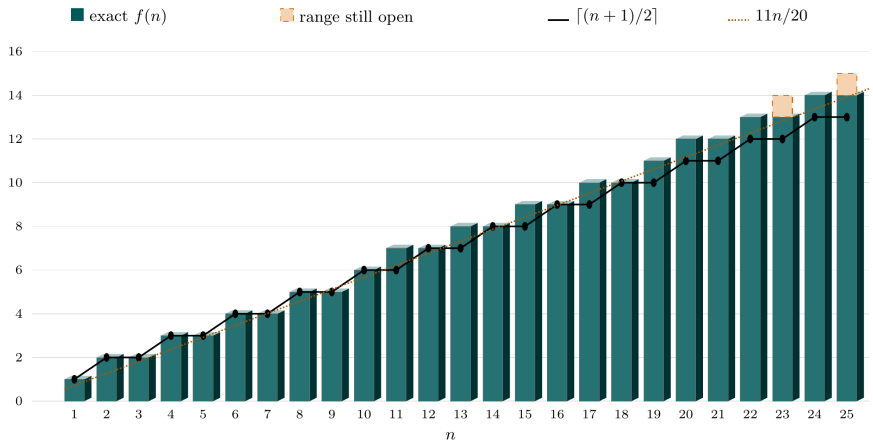


FIGURE 1. The values of $f(n)$ for $n \le 25$. The bars are exact except at $n = 23$ and $n = 25$, where $f(n)$ is one of two values. The solid line is the chain bound and the dotted line the bound $11n/20$ of [BMB26b].

$f(21) = 12$, $f(22) = 13$ *and* $f(24) = 14$. *Moreover* $13 \le f(23) \le 14$ *and* $14 \le f(25) \le 15$.

Corollary 1.5. *The chain bound is exact,* $f(n) = \lceil (n+1)/2 \rceil$*, if and only if* $n \in \{1, 2, \ldots, 10\} \cup \{12, 14, 16, 18\}$.

Proof. For the listed n this is Theorem 1.4. For $n = 20$ and $n = 22$ Theorem 1.4 gives a larger value. For odd $n \ge 11$ and for all $n \ge 23$ the lower bound $f(n) \ge \lceil (n + 1 + q)/2 \rceil$ with $q = \lfloor (n+1)/12 \rfloor$ of [BMB26b, Theorem 1.4] exceeds $\lceil (n+1)/2 \rceil$. □

The values are shown in Figure 1. The upper bounds in Theorems 1.3 and 1.4 are given by explicit words (Table 1, Figure 5); the lower bounds for $f_{\mathrm{card}}(n)$ and the absence of large monochromatic sublattices under the words are finite statements, which were checked by computer (Section 6). For $n \le 10$ the word g_n attains the chain bound, so the lower bound there is (1.1); for the other $n \le 24$ the lower bounds for $f(n)$ are those of [BMB26b]. The value $f(8) = 5$ was stated earlier as an unchecked candidate [Nd26], and the bound $f(11) \ge 7$, which refutes the equality $f(n) = \lceil (n+1)/2 \rceil$ asked about in the first version of [BMB26a], was proved in [OR26]; we know of no other exact values in the literature (searched 14 October 2026).

Two features of the data stand out. First, the best colourings found for $n \le 24$ are all cardinality colourings, and wherever $f(n)$ is known it equals $f_{\mathrm{card}}(n)$. Second, $f_{\mathrm{card}}(25) = 15$ while the best lower bound for $f(25)$ is 14; either a better lower bound for $f(25)$ exists or the optimal colourings

stop being cardinality colourings at $n = 25$. The same alternative holds at $n = 23$, where $f_{\mathrm{card}}(23) = 14$ and $f(23) \geq 13$.

Organisation. Section 2 recalls the description of sublattices by posets. Section 3 counts posets by their number of down-sets and Section 4 proves Theorem 1.1. Section 5 turns cardinality colourings into a problem about weighted posets, and Section 6 proves Theorems 1.3 and 1.4. Section 7 lists open questions. The product-of-chains construction of Section 4 and a set-theoretic form of the growth identity of Lemma 3.1 are also proved in Lean 4 [MU21] with Mathlib [Mat20].

Notation. Logarithms are to base 2 unless a base is written. For a finite poset J, a *down-set* is a subset $D \subseteq J$ such that $y \in D$ whenever $y \leq x \in D$, and $d(J)$ is the number of down-sets of J. The *width* of J is the largest size of an antichain. A poset on $[h]$ is *naturally labelled* if $i <_J j$ implies $i < j$.

2. Sublattices and posets

The following is Birkhoff's representation of finite distributive lattices [Bi37], written for sublattices of $2^{[n]}$; a form of it is [BMB26b, Lemma 7.8]. We include the short proof because the sizes of the members are what we need.

Lemma 2.1. *Let $\mathcal{L}$ be a sublattice of $2^{[n]}$, let b be its least and t its largest member, and for $x \in t \setminus b$ let L_x be the least member of $\mathcal{L}$ that contains x. Put $x \preceq y$ if $L_x \subseteq L_y$, let J be the poset of classes of the preorder $\preceq$, and let $\varphi\colon t \setminus b \to J$ send x to its class. Then*

$$D \;\longmapsto\; b \cup \varphi^{-1}(D)$$

is a bijection from the down-sets of J onto $\mathcal{L}$, and $|b \cup \varphi^{-1}(D)| = |b| + w(D)$, where $w(j) = |\varphi^{-1}(j)| \geq 1$ and $w(D) = \sum_{j \in D} w(j)$. Conversely, for disjoint $b, X \subseteq [n]$, a finite poset J and a surjection $\varphi\colon X \to J$, the family $\{b \cup \varphi^{-1}(D)\}$, D a down-set of J, is a sublattice with $d(J)$ members.

Proof. Since $\mathcal{L}$ is closed under intersection, L_x is the intersection of the members that contain x, and every member contains b. If $y \in L_x$, then L_x is a member containing y, so $L_y \subseteq L_x$. Let $A \in \mathcal{L}$ and $X = A \setminus b$. For $x \in X$ we have $L_x \subseteq A$, so $A = b \cup \bigcup_{x \in X} L_x$, and X is a union of classes forming a down-set: if $L_y \subseteq L_x$ and $x \in A$, then $y \in L_y \subseteq A$. Conversely, if D is a down-set of J and $X = \varphi^{-1}(D)$, then $B = b \cup \bigcup_{x \in X} L_x$ lies in $\mathcal{L}$, as a union of members (or $B = b$ if D is empty), and $B = b \cup X$: every $y \in L_x \setminus b$ with $x \in X$ has $L_y \subseteq L_x$, so $\varphi(y) \in D$. The two maps are inverse to each other, which proves the bijection; the sizes are clear. The converse holds because unions and intersections of down-sets are down-sets and φ^{-1} preserves them. □

A naturally labelled poset on $[h]$ exists in every isomorphism class of h-element posets, by choosing a linear extension. Recording, for each point of $[n]$, whether it lies in b, outside t, or in the class $\varphi(x) \in [h]$, we obtain the following count.

Lemma 2.2. *Let $P(h,m)$ be the number of naturally labelled posets on $[h]$ with exactly m down-sets. The number of sublattices of $2^{[n]}$ with exactly $m \geq 2$ members is at most*

$$\sum_{h=1}^{n} (h+2)^n \, P(h,m).$$

3. Posets with a given number of down-sets

Let J be a naturally labelled poset on $[h]$ and let J_j be its restriction to $[j]$. For $j \in [h]$ let

$$D_j = \{i < j : \ i <_J j\}, \qquad U_j = [j-1] \setminus D_j, \qquad u_j = d(U_j),$$

where U_j carries the order induced from J. Then D_j is a down-set and U_j an up-set of J_{j-1}, and J_j is determined by J_{j-1} and U_j (Figure 2).

Lemma 3.1. *$d(J_j) = d(J_{j-1}) + u_j$ for every j, and hence $d(J) = 1 + u_1 + \cdots + u_h$ with every $u_j \geq 1$.*

Proof. The down-sets of J_j that do not contain j are the down-sets of J_{j-1}. Those that contain j are the sets $E \cup \{j\}$ with E a down-set of J_{j-1} containing D_j, and $E \mapsto E \setminus D_j$ is a bijection from these E onto the down-sets of U_j: if F is a down-set of U_j, then $F \cup D_j$ is a down-set of J_{j-1}, because an element below a member of D_j lies in D_j, and an element below a member of F lies in D_j or in U_j, and in the latter case in F. The empty poset has one down-set. □

Two facts bound the number of choices of U_j. If A is an antichain of a poset Q, every subset of A that consists of minimal elements of Q is a down-set of Q; and every antichain of J gives $2^{|A|}$ distinct down-sets of J, namely the down-closures of its subsets, so

$$2^{\mathrm{width}(J)} \ \leq \ d(J). \tag{3.1}$$

Lemma 3.2. *Let $w = \max\{1, \mathrm{width}(J_{j-1})\}$. Given J_{j-1} and $u_j = u$, there are at most $(wu)^{\log u}$ possibilities for U_j.*

Proof. An up-set is determined by its set A of minimal elements. These form an antichain of minimal elements of U_j, so every subset of A is a down-set of U_j and $2^{|A|} \leq u$. Fix a partition of J_{j-1} into w chains, which exists by Dilworth's theorem [Di50]. Let $a \in A$ lie in the chain C, and let r be the number of elements of C above a. They lie in the up-set U_j, so U_j contains a chain $a = c_0 < c_1 < \cdots < c_r$, whose down-closures in U_j together with $\varnothing$

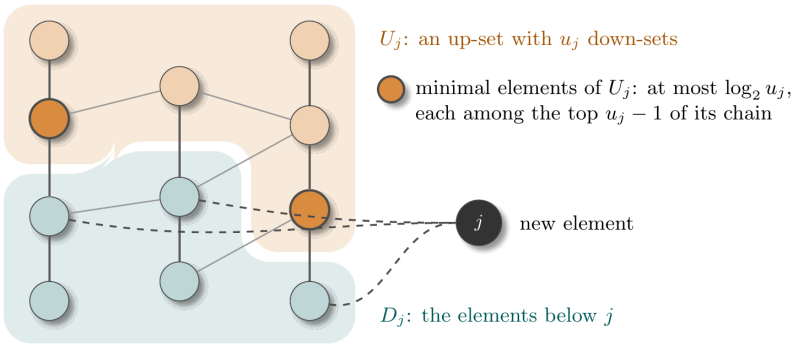


FIGURE 2. Adding the element j to J_{j-1}. The down-sets of J_j that contain j correspond to the down-sets of the up-set U_j, and U_j is determined by its minimal elements, which sit near the tops of the chains of a chain partition.

are $r+2$ distinct down-sets of U_j. Hence $r \le u-2$, and a is one of the $u-1$ largest elements of its chain. So A is a set of at most $\log u$ elements chosen among at most $N = w(u-1)$ candidates, and the number of choices is at most $\sum_{i \le \log u} \binom{N}{i} \le (N+1)^{\log u} \le (wu)^{\log u}$. □

Proposition 3.3. *For $h \ge 1$ and $m \ge 2$,*

$$\log P(h,m) \;\le\; \log \binom{m-2}{h-1} \;+\; \max \sum_{j=1}^{h} \log u_j \,\big(\log u_j + \log\log m\big),$$

the maximum taken over $u_1, \dots, u_h \ge 1$ with $u_1 + \cdots + u_h = m-1$. Consequently, for all $\delta, \gamma > 0$ there is n_0 such that

$$P(h,m) \;\le\; 2^{\delta m} \qquad \text{whenever } n \ge n_0,\ 1 \le h \le n \text{ and } m \ge \gamma\, n \log n.$$

Proof. By Lemma 3.1 the sequence $(u_1, \dots, u_h)$ is a composition of $m-1$ into h positive parts, and there are $\binom{m-2}{h-1}$ of them. Given the sequence, the poset is built one element at a time, and Lemma 3.2 bounds the number of choices at step j by $(wu_j)^{\log u_j}$ with $w \le \max\{1, \log m\}$ by (3.1), because $\mathrm{width}(J_{j-1}) \le \mathrm{width}(J)$. This is the first inequality.

For the second, we may assume $m \le 2^h \le 2^n$, since a poset on $[h]$ has at most 2^h down-sets, and that n is so large that $\gamma \log n \ge e$. Write $x = m/h \ge m/n \ge \gamma \log n$. The function $\psi(y) = (\log(ey))^2$ is concave and increasing on $[1, \infty)$ and $\psi(y) \ge (\log y)^2$ there, and log is concave, so by Jensen's inequality

$$\sum_j (\log u_j)^2 \le h\, \psi(m/h) = m\, \frac{\psi(x)}{x}, \qquad \sum_j \log u_j \le h \log(m/h) = m\, \frac{\log x}{x}.$$

The functions $\psi(y)/y$ and $(\log y)/y$ both decrease for $y \geq e$, so that

$$\frac{\psi(x)}{x} \leq \frac{\psi(\gamma \log n)}{\gamma \log n} \longrightarrow 0.$$

For the term with $\log\log m$ we use $\log\log m \leq \log n$ and distinguish two cases. If $m \leq n^3$, then $\log\log m \leq \log(3\log n)$ and

$$\frac{\log x}{x} \leq \frac{\log(\gamma \log n)}{\gamma \log n};$$

if $m > n^3$, then $x > n^2$ and $(\log x)/x < 2\log n/n^2$. In both cases $\log\log m \cdot (\log x)/x \to 0$. Finally,

$$\log \binom{m-2}{h-1} \leq h \log \frac{em}{h} = m\,\frac{\log(ex)}{x}, \qquad \frac{\log(ex)}{x} \leq \frac{\log(e\gamma \log n)}{\gamma \log n} \to 0.$$

All three ratios to m tend to 0 uniformly in $h \leq n$ and $m \geq \gamma\, n \log n$, which proves the second statement. □

Remark 3.4. The bound of Proposition 3.3 is far from the number $2^{h^2/4+O(h)}$ of all posets on $[h]$ [KR75]. The point is that posets on $h \leq n$ elements with $m \geq n\log n$ down-sets are thin: their width is at most $\log m$, and each new element is attached by at most $\log u_j$ generators near the tops of the chains.

4. Random colourings

4.1. **The upper bound.** Let χ be a uniformly random k-colouring of $2^{[n]}$, let $\delta > 0$ and put $m = \lceil (1+\delta)\, n \log_k n \rceil$. A fixed family of M distinct sets is monochromatic with probability k^{1-M}. If some monochromatic sublattice has at least m members, then some sublattice with exactly $M \geq m$ members is monochromatic; we do not need to pass to a sub-sublattice of exactly m members, but sum over all $M \geq m$. By Lemma 2.2 and Proposition 3.3, applied with $\gamma = 1/\log k$ and a constant $\eta > 0$ in place of δ to be chosen, for large n,

$$\begin{aligned} \mathbb{P}[\Lambda(\chi) \geq m] \;&\leq\; \sum_{M \geq m} k^{1-M} \sum_{h=1}^{n} (h+2)^n P(h, M) \\ &\leq\; \sum_{M \geq m} kn\,(n+2)^n\, 2^{\eta M} k^{-M}. \end{aligned}$$

Since $(n+2)^n \leq e^2 n^n$ and $n\log n = \log k \cdot n \log_k n \leq M \log k/(1+\delta)$ for $M \geq m$, the logarithm of the summand is at most

$$\log(e^2 kn) \;-\; M\Big(\log k - \eta - \frac{\log k}{1+\delta}\Big) \;=\; \log(e^2 kn) - cM,$$

where $c = \log k \cdot \delta/(1+\delta) - \eta$. Taking $\eta = \log k \cdot \delta/(2+2\delta)$ makes $c > 0$, and then

$$\mathbb{P}[\Lambda(\chi) \geq m] \;\leq\; \frac{e^2 kn\, 2^{-cm}}{1 - 2^{-c}} \;\longrightarrow\; 0,$$

because $m \geq n$. This proves the upper bound of Theorem 1.1 and, since a colouring with $\Lambda(\chi) < m$ exists for large n, Corollary 1.2.

4.2. **The lower bound.** Let $0 < \varepsilon < 1$, put $\mu = \lfloor (1-\varepsilon)\log_k n \rfloor$ and assume n is so large that $\mu \geq 1$. Let $Z = \{z_1, \dots, z_{\mu-1}\}$ consist of the last $\mu - 1$ points of $[n]$, let $\mathcal{M}$ be the chain $\varnothing \subset \{z_1\} \subset \{z_1, z_2\} \subset \cdots \subset Z$ with μ members, and let $Y = [n] \setminus Z$. Fix a colour c. Call a set $A \subseteq Y$ *good* if $A \cup S$ has colour c for every $S \in \mathcal{M}$. Distinct sets A are tested on disjoint collections of sets $A \cup S$, since $A = (A \cup S) \cap Y$; hence the events that different sets are good are independent, each of probability

$$p = k^{-\mu} \;\geq\; n^{-(1-\varepsilon)}.$$

We build a chain $A_1 \subset A_2 \subset \dots$ of good sets with $|A_i| = i$ greedily. Put $A_0 = \varnothing$. At step $i \geq 1$ we examine the $|Y| - i + 1$ sets $A_{i-1} \cup \{y\}$ with $y \in Y \setminus A_{i-1}$ and let A_i be the first good one. Every set examined at step i has size i, so it was not examined before, and its goodness is independent of everything revealed so far. Hence step i fails with probability

$$(1-p)^{|Y|-i+1} \;\leq\; \exp\bigl(-p(|Y| - i + 1)\bigr),$$

which is at most $\exp(-n^{\varepsilon/2})$ as long as $|Y| - i + 1 \geq n^{1-\varepsilon/2}$. By the union bound, with probability at least $1 - n\exp(-n^{\varepsilon/2})$ the procedure produces good sets $A_1 \subset \cdots \subset A_h$ with $h \geq |Y| - n^{1-\varepsilon/2} \geq n - \log_k n - n^{1-\varepsilon/2}$.

The family $\{A_i \cup S : \; 1 \leq i \leq h, \; S \in \mathcal{M}\}$ is a sublattice, since for chains in disjoint ground sets

$$(A \cup S) \cup (A' \cup S') = (A \cup A') \cup (S \cup S'),$$
$$(A \cup S) \cap (A' \cup S') = (A \cap A') \cup (S \cap S'),$$

and the union and the intersection of two members of a chain is one of them. Its members are distinct, again because $A = (A \cup S) \cap Y$, and all have colour c. It has

$$h\mu \;\geq\; \bigl(n - \log_k n - n^{1-\varepsilon/2}\bigr)\bigl((1-\varepsilon)\log_k n - 1\bigr) \;\geq\; (1-2\varepsilon)\, n \log_k n$$

members for large n. Applying this to each of the k colours and replacing ε by $\varepsilon/2$ proves the lower bound of Theorem 1.1, including the statement for every colour. □

Remark 4.1. The two halves of the proof explain the constant. A sublattice with m members in which every set is a product of a long chain and a chain of length μ uses about n independent tests, each of probability $k^{-\mu}$; the long chain is available only while $k^{\mu} \leq n$. The counting bound says that no other shape does better to first order, since there are only $(n+2)^n 2^{o(m)} = k^{(1+o(1))\, n \log_k n}$ sublattices with $m \approx n \log_k n$ members.

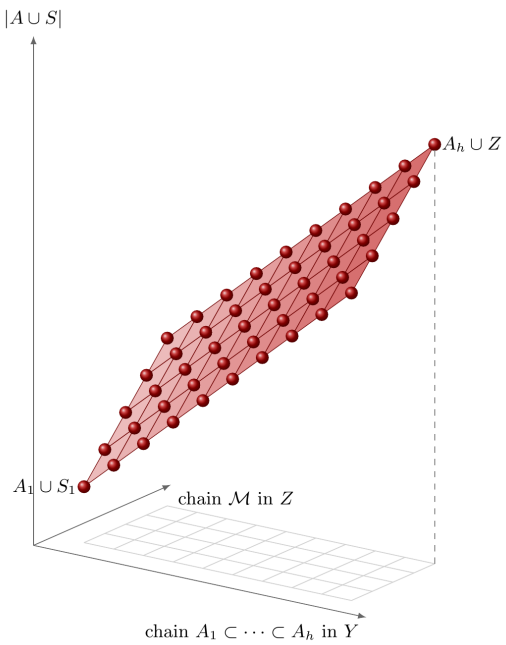


FIGURE 3. The monochromatic sublattice built in the lower bound: a greedy chain of good sets in Y times a short chain in Z. Its members form a grid, drawn with height equal to the size of each set.

5. Colourings by cardinality

Let $g\colon \{0,1,\dots,n\} \to \{\mathsf{R},\mathsf{B}\}$ and colour $A \subseteq [n]$ by $g(|A|)$. For a colour c let $S_c = g^{-1}(c)$. A *weighted poset* is a finite poset J with a weight $w\colon J \to \{1,2,\dots\}$, and a *placement* of it in $\{0,\dots,n\}$ is an integer $\beta \geq 0$ with $\beta + w(J) \leq n$. By Lemma 2.1 the sublattices of $2^{[n]}$ are, up to the choice of the points, the placed weighted posets, and a member corresponding to a down-set D has size $\beta + w(D)$. Hence:

Lemma 5.1. *The largest monochromatic sublattice of colour c under g has as many members as the largest $d(J)$ over placed weighted posets (J, w, β) with $\beta + w(D) \in S_c$ for every down-set D of J.*

We call such a placed weighted poset *c-monochromatic*. In Figure 4 the poset $a \parallel (b < c)$ with weights $1, 3, 5$ and $\beta = 0$ has down-set sums $0, 1, 3, 4, 8, 9$, all red for the word g_{10} of Table 1; this red sublattice has six members, which is the largest possible for that word.

Two properties make the search over weighted posets finite and complete.

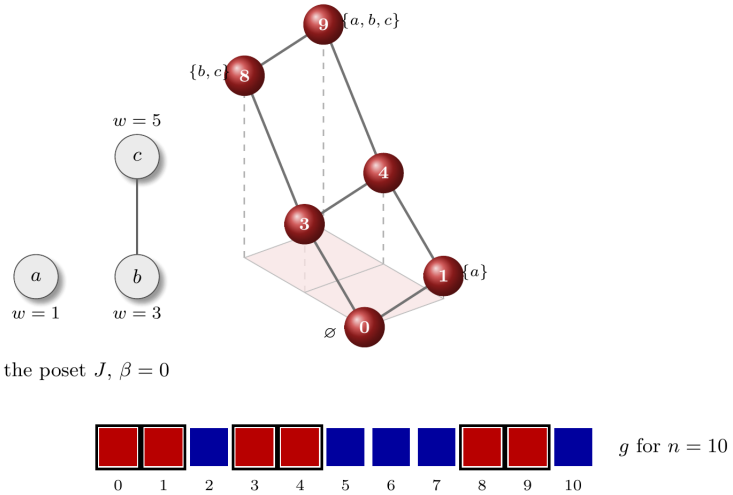


FIGURE 4. A weighted poset, its lattice of down-sets with the down-set sums, and the word g_{10}, which is red at all six sums.

Lemma 5.2. *Let (J, w, β) be c-monochromatic, with J naturally labelled on $[h]$. Then each restriction $(J_j, w|_{[j]}, \beta)$ is c-monochromatic, and if $d(J) \geq K$ there is j with $K \leq d(J_j) \leq K - 1 + u_j$.*

Proof. A down-set of J_j is a down-set of J with the same sum. The numbers $d(J_j)$ increase from $d(J_0) = 1$ to $d(J)$ in steps u_j by Lemma 3.1; take the first j with $d(J_j) \geq K$. □

So to decide whether g has a c-monochromatic sublattice with K members it suffices to grow naturally labelled weighted posets one element at a time, attaching the new element above a down-set D_j of the current poset with a weight $w_j \geq 1$, keeping only those placements in which all down-set sums lie in S_c, and stopping as soon as K down-sets are reached. Each attachment adds the sums $\beta + w(E) + w_j$ for the down-sets $E \supseteq D_j$, so the condition can be checked incrementally. The dual procedure, which adds minimal elements below an up-set and keeps the top value $\beta + w(J)$ fixed, gives a second, independent decision procedure.

Arithmetic progressions in S_c give Boolean cubes: if g is constant on $a, a + e, \ldots, a + (p - 1)e$, then g has a monochromatic sublattice with 2^{p-1} members [BMB26b, Proposition 6.2]. Thus a word without large monochromatic sublattices has no long monochromatic progressions, and the words of Table 1 have none of length five; for $n \geq 16$ their longest monochromatic progressions have length four.

n	$f_{\mathrm{card}}(n)$	word g_n	n	$f_{\mathrm{card}}(n)$	word g_n
1	1	RB	14	8	RBRRBBBRRBBRBRR
2	2	RRB	15	9	RBBRBRRBBBRBRRRB
3	2	RRBB	16	9	RRBRBBRRBBBRBRRBR
4	3	RRBRB	17	10	BRBBRRBBBRBRRBBRRR
5	3	RRBRBB	18	10	RRBBRRBRBBBRRBBRBRR
6	4	BBRRBRR	19	11	BRBBRRRBBRBRRBBBRRBR
7	4	BBRRBBRR	20	12	RRBBRBBBRRBRBBRRRBBRR
8	5	RBBBRRBRR	21	12	BBRRBBRBRRRBBBRBRRBBRR
9	5	BBRBRRBBRR	22	13	BRRBRBBRRRBBRRBRBBBRRBR
10	6	RRBRRBBBRRB	23	14	BBRRBRBBBRRBRBBRRRBBRRBR
11	7	RRRBRRBBBRBB	24	14	RRBBRBRRBBBRBRRBBRBRBRRBR
12	7	RRBBBRRBRBBRR	25	15	BRBBRRBBBRBRRBRRRBBBRBRRBR
13	8	RBBBRRBRBBRRRB			

TABLE 1. Optimal cardinality colourings. Under g_n every monochromatic sublattice of $2^{[n]}$ has at most $f_{\mathrm{card}}(n)$ members.

6. Small values

Lemma 6.1. *For $1 \le n \le 25$, every monochromatic sublattice of $2^{[n]}$ under the cardinality colouring g_n of Table 1 has at most $f_{\mathrm{card}}(n)$ members, with $f_{\mathrm{card}}(n)$ as in Table 1.*

Proof. By Lemma 5.1 it suffices to show that for each colour c there is no c-monochromatic placed weighted poset with $f_{\mathrm{card}}(n)+1$ down-sets, and by Lemma 5.2 the search described after it decides this. Both the search and its dual were run for every $n \le 25$ and both colours, and both found no such poset. □

Lemma 6.2. *For $1 \le n \le 25$, every word $g\colon \{0,\dots,n\} \to \{\mathsf{R},\mathsf{B}\}$ has a monochromatic sublattice with at least $f_{\mathrm{card}}(n)$ members, with $f_{\mathrm{card}}(n)$ as in Table 1.*

Proof. For $n \le 10$ this is the chain bound (1.1). For the other n we produced a list $\mathcal{W}_n$ of placed weighted posets, each with at least $f_{\mathrm{card}}(n)$ down-sets, and checked that every word is constant on the set of down-set sums of some member of $\mathcal{W}_n$; by Lemma 5.1 this proves the claim. Each member of $\mathcal{W}_n$ is a pair of clauses of a propositional formula in the variables $g(0),\dots,g(n)$, saying that g is not constant on its set of sums, and the claim is that the formula is unsatisfiable. The lists were found by alternating a satisfiability solver, which proposes a word, with the search of Section 5, which returns a monochromatic poset for that word. Each certificate was then checked independently: the down-sets of every poset in $\mathcal{W}_n$ were recomputed from scratch, and the unsatisfiability of the formula was confirmed by two different

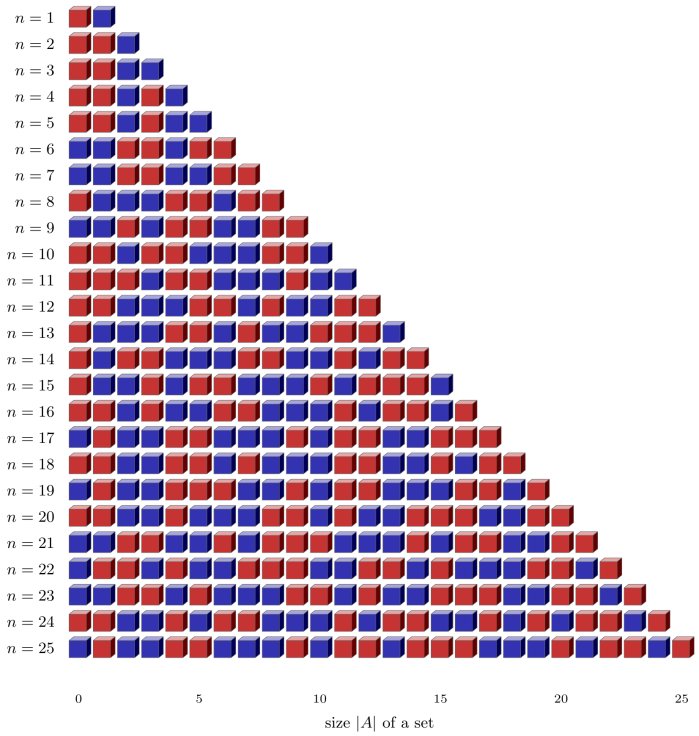


FIGURE 5. The words g_n of Table 1; the cell in column i of row n is the colour of the sets of size i.

solvers. The lists have between 288 (for $n = 11$) and 49,078 (for $n = 25$) members. □

Proof of Theorem 1.3. Lemmas 6.1 and 6.2. □

Proof of Theorem 1.4. The upper bounds $f(n) \le f_{\mathrm{card}}(n)$ hold because cardinality colourings are colourings. For the lower bounds let $n+1 = 20q + r$ with $0 \le r \le 19$ and $e = [r \ge 12]$, and $q' = \lfloor (n+1)/12 \rfloor$. By [BMB26b, Theorems 1.4 and 7.6],

$$f(n) \;\ge\; \max\left\{ \left\lceil \frac{n+1+q'}{2} \right\rceil,\; \left\lceil \frac{n+1+2q+e}{2} \right\rceil \right\}.$$

For $n \le 10$ this is $\lceil (n+1)/2 \rceil = f_{\mathrm{card}}(n)$. For odd n with $11 \le n \le 21$ the first term is $(n+3)/2 = f_{\mathrm{card}}(n)$; for $n \in \{12, 14, 16, 18\}$ it is $(n+2)/2 = f_{\mathrm{card}}(n)$. For $n = 20$ and $n = 22$ the second term gives 12 and 13, and for $n = 24$ the first gives 14; these equal $f_{\mathrm{card}}(n)$. For $n = 23$ and $n = 25$ both terms are at most 13 and 14 respectively. □

The values $f(20) = 12$ and $f(22) = 13$ depend on the window lemma of [BMB26b, Lemma 7.7], which is proved by a finite computation; all other lower bounds in Theorem 1.4 are proved by hand in [BMB26b].

7. Questions

Theorem 1.1 locates the random colouring precisely, and (1.2) leaves a factor of order $\log n$ for $f(n)$.

Question 7.1. *Is $f_{\mathrm{card}}(n) = O(n)$? If so, is $f_{\mathrm{card}}(n)/n$ convergent?*

For $10 \le n \le 25$ the ratio $f_{\mathrm{card}}(n)/(n+1)$ lies between $1/2$ and 0.59. A positive answer would give $f(n) = O(n)$, answering [BMB26b, Question 9.3]. Good words must avoid long monochromatic progressions, since a progression of length p gives a sublattice with 2^{p-1} members, and more generally they must avoid the down-set sums of every weighted poset with many down-sets.

Question 7.2. *Is $f(23) = 13$ or 14, and is $f(25) = 14$ or 15? Is there any n with $f(n) < f_{\mathrm{card}}(n)$?*

Question 7.3. *Is $f_k(n) = (1 + o(1))\,(n+1)/k$ for $k \ge 3$, or does $f_k(n)/n$ stay above $1/k$ by a constant, as (1.2) shows for $k = 2$?*

Data availability

The words of Table 1, the certificates of Lemma 6.2, the programs that produce and check them (in C++ and Python), and the Lean formalisation are available at `https://github.com/creelie/D7`. The programs and the Lean formalisation were written with the assistance of Claude Code (Anthropic).

References

[BMB26a] D. Bhattacharjee, P. Mandal and U. Bhattacharya, *Resolving Erdős–Ulam monochromatic union-closed family conjectures*, preprint (2026), arXiv:2610.02833.

[BMB26b] D. Bhattacharjee, P. Mandal and U. Bhattacharya, *Resolving Erdős–Ulam monochromatic union-closed family conjectures*, version 5 with Lean formalisation, Zenodo (2026), doi:10.5281/zenodo.23374973.

[Bi37] G. Birkhoff, *Rings of sets*, Duke Math. J. **3** (1937), no. 3, 443–454. doi:10.1215/S0012-7094-37-00334-X

[Bl26] T. F. Bloom, *Erdős Problem #1183*, `https://www.erdosproblems.com/1183`, accessed 14 October 2026.

[Ch26] P. Chojecki, *Monochromatic union-closed and lattice subfamilies of the Boolean lattice*, preprint, 18 March 2026, `https://www.ulam.ai/research/erdos1183.pdf`.

[Di50] R. P. Dilworth, *A decomposition theorem for partially ordered sets*, Ann. of Math. (2) **51** (1950), no. 1, 161–166. doi:10.2307/1969503

[KR75] D. J. Kleitman and B. L. Rothschild, *Asymptotic enumeration of partial orders on a finite set*, Trans. Amer. Math. Soc. **205** (1975), 205–220. doi:10.1090/S0002-9947-1975-0369090-9

[Mat20] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and

Proofs, CPP 2020, ACM, New York, 2020, 367–381. doi:10.1145/3372885.3373824

[MU21] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction, CADE 28, Lecture Notes in Comput. Sci. 12699, Springer, Cham, 2021, 625–635. doi:10.1007/978-3-030-79876-5_37

[Nd26] ndrorchestration, *Stage Erdős #1183* $f(8) = 5$ *and* $F(8) \leq 8$ *candidate verification*, GitHub pull request, 15 September 2026, https://github.com/ndrorchestration/ndrorchestration/pull/19.

[OR26] The Omega Institute, *Refute* $f(n) = \lceil (n+1)/2 \rceil$ *via* $f(11) \geq 7$, trureturing issue #14535, 8–9 October 2026, https://github.com/the-omega-institute/trureturing/issues/14535.

Formerly, Electro-Gravitational Space Propulsion Laboratory (EGSPL), Bhubaneswar, Odisha 751030, India
Email address: d.bhattacharjee@erl-forschung.de
Email address: itsdeep@live.com
ORCID: 0000-0003-0466-750X